

PARTICLE-ONLY DLR EQUATIONS AND STATIONARY UNIQUENESS FOR ONE-DIMENSIONAL COULOMB JELLIUM

ABSTRACT. We specify canonical particle-only DLR equations for the genuine one-dimensional Coulomb potential $g(x) = -|x|$, using symmetric Cesàro averages of particle-background energy differences and an explicit summability domain. At every inverse temperature $\beta > 0$ and background density $\rho > 0$, these equations have exactly one point-process solution of intensity ρ invariant under all real translations. The principal step is a more general lifting theorem: finite Gaussian conditional densities with the Coulomb curvature determine a measurable stationary electric field through consistency on nested intervals. The exterior field contains exactly the information in the exterior particles and the interior count. No discrepancy bound, field moment, or deterministic asymptotic density is assumed in this theorem. A complete entropy and Palm proof of stationary field-DLR uniqueness is included. Existence is deduced from published finite-volume field estimates. The argument does not require a classification of all nonstationary crystalline phases.

1. INTRODUCTION AND MAIN CONCLUSIONS

The potential considered throughout is

$$g(x) = -|x|, \quad -g'' = 2\delta_0,$$

with background charge $-\rho dx$ and particles of charge $+1$. This is not the logarithmic interaction, nor the three-dimensional Coulomb potential restricted to a line. Our pair-background energy includes a factor $1/2$ in the double integral; its field representation is $\int E^2$ when $E' = \gamma - \rho dx$.

Long-range interactions require a summation convention before an infinite-volume DLR statement has a definite meaning. Section 3 defines such a convention directly from finite particle-background energies. It does not put existence of a field lift into the definition.

Theorem 1.1 (Stationary particle uniqueness). *For every $\beta, \rho > 0$, the symmetric Cesàro canonical specification of Definition 3.3 has exactly one probability solution $P_{\beta, \rho}$ on $\text{Conf}(\mathbb{R})$ which has intensity ρ and is invariant under all real translations. Every such solution admits the deterministic jointly stationary field-DLR lift*

$$Q(d\gamma, dc) = P_{\beta, \rho}(d\gamma)\delta_{-L_\rho(\gamma)}(dc),$$

where L_ρ is the particle statistic in (3.1). Its almost-sure density in each direction is ρ .

There are three separate ingredients. The lifting theorem in Section 4 applies to any particle law with finite, exterior-measurable Gaussian conditional coefficients. It uses neither Cesàro summability nor a field moment. Sections 5–8 prove at most one stationary field-DLR law by a finite-block entropy comparison. Section 9 supplies existence for the specified normalization using established finite-volume estimates.

The distinction between a prescribed mean intensity and an almost-sure density is essential: the latter is derived, including for nonergodic laws. Empty configurations require separate treatment because canonical resampling can never insert the first particle. They are explicitly accounted for in Theorem 4.1.

The classical works of Kunz [4] and Aizenman–Martin [1] are the relevant historical background. Aizenman–Jansen–Jung [2] give an accessible field formulation and the estimates used here. Our claim is a theorem for the standalone specification stated below; we make no claim of priority for stationary uniqueness and do not use a classical classification theorem with an unspecified admissibility class.

2. CONFIGURATION SPACE AND FIELD KERNELS

Let $\text{Conf}(\mathbb{R})$ be the space of locally finite simple counting measures, with its usual Borel structure. Write $N_B(\gamma) = \gamma(B)$ and $\theta_t \gamma = \gamma - t$. A stationary law of finite intensity λ satisfies $\mathbb{E}N_B = \lambda|B|$ for bounded Borel B . In particular, every deterministic location is unoccupied almost surely. The background parameter ρ need not initially equal λ .

For any configuration define the right-continuous anchored primitive

$$D_\gamma(t) = \begin{cases} N_{(0,t]}(\gamma) - \rho t, & t \geq 0, \\ -N_{(t,0]}(\gamma) - \rho t, & t < 0. \end{cases} \quad (2.1)$$

Then $D_\gamma(0) = 0$, $D'_\gamma = \gamma - \rho dx$, and, for $s < t$,

$$D_\gamma(t) - D_\gamma(s) = N_{(s,t]} - \rho(t - s). \quad (2.2)$$

All finite-valued right-continuous fields with this derivative have the form $E_\gamma(x) = C + D_\gamma(x)$.

For $I = (a, b)$ let

$$\mathcal{F}_I = \sigma(\gamma_{I^c}, N_I).$$

This is the conditioning sigma-field for a canonical particle kernel. For $n \geq 0$ and $e \in \mathbb{R}$, let $K_{I,n,e}$ be the probability on ordered points in I with density

$$\frac{1}{Z_{I,n,e}} \exp \left\{ -\beta \rho \sum_{i=1}^n \left(x_i - a - \frac{e + i - \frac{1}{2}}{\rho} \right)^2 \right\} \mathbf{1}_{\{a < x_1 < \dots < x_n < b\}} dx_1 \cdots dx_n. \quad (2.3)$$

For $n = 0$ this is the point mass on the empty configuration. The normalizing constant is finite and positive for every finite e, n .

Definition 2.1 (Canonical field-DLR). A law of (γ, C) , with $E = C + D_\gamma$, satisfies field-DLR if, conditionally on the particles and field outside each bounded deterministic interval $I = (a, b)$, the interior particles have law $K_{I,N_I,E(a)}$. The exterior field is retained and the interior field is

$$E(t) = E(a) + \sum_{i=1}^{N_I} \mathbf{1}_{\{x_i \leq t\}} - \rho(t - a), \quad a < t < b. \quad (2.4)$$

For stationary finite-intensity laws, endpoint particles have probability zero, and the exterior field determines $N_I = E(b) - E(a) + \rho(b - a)$. If $0 \in I$, C is updated to the value of the resampled field at zero.

Translations of fields are $E(x) \mapsto E(x + t)$. Joint stationarity is equivalent to invariance in law under

$$(\gamma, C) \mapsto (\theta_t \gamma, C + D_\gamma(t)).$$

Equality of this law for each fixed real t is the stationarity requirement; a single exceptional null set valid for all t is not needed.

3. A PARTICLE-ONLY CESÀRO SPECIFICATION

Definition 3.1 (Summability domain). Let $\mathcal{A}_\rho$ consist of configurations for which the following limit exists and is finite:

$$L_\rho(\gamma) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T D_\gamma(u) du. \quad (3.1)$$

This is a Borel set: the finite- T expressions are Borel in γ and continuous in $T > 0$, so rational T suffice to test the limit. No discrepancy bound or asymptotic density is part of the definition. The cancellation of the odd function ρu also shows that the set and statistic in (3.1) are independent of ρ ; the background parameter still enters all local energies.

For $I = (a, b)$, an exterior configuration ζ , and an interior configuration ξ , write

$$\mu_R^\xi = (\xi + \zeta - \rho dx)|_{(-R, R]}, \quad U_R(\xi) = \frac{1}{2} \iint g(x - y) d\mu_R^\xi(x) d\mu_R^\xi(y).$$

Endpoint conventions for R do not change any of the averages below. There is no point self-energy divergence since $g(0) = 0$. For two interiors ξ, η of equal cardinality define

$$\Delta H_I(\xi, \eta; \zeta) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{R_0}^T (U_R(\xi) - U_R(\eta)) dR, \quad (3.2)$$

where $(-R_0, R_0)$ contains $\bar{I}$.

Proposition 3.2 (Finite local energies and canonical closure). *Suppose $\eta + \zeta \in \mathcal{A}_\rho$ and $|\xi| = |\eta|$. Then $\xi + \zeta \in \mathcal{A}_\rho$, (3.2) exists, and, on writing*

$$\mathcal{E}_\gamma(x) = D_\gamma(x) - L_\rho(\gamma), \quad (3.3)$$

one has

$$\mathcal{E}_{\xi+\zeta} = \mathcal{E}_{\eta+\zeta} \quad \text{outside } I, \quad (3.4)$$

$$\Delta H_I(\xi, \eta; \zeta) = \int_a^b (\mathcal{E}_{\xi+\zeta}^2 - \mathcal{E}_{\eta+\zeta}^2) dt. \quad (3.5)$$

The relative energy is independent of R_0 and satisfies the cocycle identity in its two interior arguments.

Proof. Let $\delta = \xi - \eta$ and let $d(t) = \delta((-\infty, t])$. Since $\delta(\mathbb{R}) = 0$, d vanishes outside I . The anchored primitives satisfy

$$D_{\xi+\zeta}(t) - D_{\eta+\zeta}(t) = d(t) - d(0).$$

Taking symmetric means gives $L_\rho(\xi + \zeta) - L_\rho(\eta + \zeta) = -d(0)$. This proves canonical closure and $\mathcal{E}_{\xi+\zeta} - \mathcal{E}_{\eta+\zeta} = d$.

For $|t| < R$, the right-continuous field $-\frac{1}{2}(g * \mu_R^\eta)'$ is

$$E_R^\eta(t) = D_{\eta+\zeta}(t) - \frac{D_{\eta+\zeta}(R) + D_{\eta+\zeta}(-R)}{2}. \quad (3.6)$$

The integration-by-parts identity for the neutral compactly supported measure $\delta = d'$ gives

$$U_R(\xi) - U_R(\eta) = \int_I (2E_R^\eta(t)d(t) + d(t)^2) dt.$$

No neutrality assumption on μ_R^η is necessary: d is compactly supported, so no boundary term remains. The Cesàro mean of the last term in (3.6) is $L_\rho(\eta + \zeta)$, which proves (3.5). The remaining statements follow directly. $\square$

Definition 3.3 (Symmetric Cesàro canonical particle-DLR). For $\gamma \in \mathcal{A}_\rho$, retain $\zeta = \gamma_{I^c}$ and $n = N_I$. The conditional density of n ordered interior points is proportional to

$$\exp[-\beta \Delta H_I(\xi, \eta; \zeta)] \mathbf{1}_{\{a < x_1 < \dots < x_n < b\}} dx_1 \cdots dx_n, \quad (3.7)$$

where η is any n -point completion in I . A probability P satisfies these DLR equations if $P(\mathcal{A}_\rho) = 1$ and (3.7) is its conditional law given $\mathcal{F}_I$ for every bounded deterministic interval I .

The definition is standalone and uses only particle-background energies. Formula (3.3) is a derived statistic, not an assumption about a jointly stationary extension.

For explicit measurability, choose the reference completion $\eta = \sum_{i=1}^n \delta_{a+i(b-a)/(n+1)}$. Canonical closure shows that admissibility can be tested on this completion. The parameter

$$e_I(\zeta, n) = \mathcal{E}_{\eta+\zeta}(a) \quad (3.8)$$

is therefore a Borel function of the retained data. It agrees with $\mathcal{E}_\gamma(a)$ for any admissible completion. Moving the i th ordered jump gives

$$\partial_{x_i} \int_I \mathcal{E}^2 = 2\rho(x_i - a) - 2e_I - 2i + 1.$$

Thus (3.7) is exactly K_{I,n,e_I} . In particular, its normalization is finite and positive. The kernels retain admissibility and are consistent under nesting: their exterior field is fixed by (3.4), and the integral energy splits over subintervals. These statements also follow by conditioning the explicit ordered Gaussian density.

Remark 3.4 (Scope of the normalization). On arbitrary configurations $\mathcal{A}_\rho$ need not be invariant under translation, and the normalized primitive need not be covariant. We assert neither property. They hold almost surely in the sense needed for stationary candidate laws by Theorem 4.1 below. The zero symmetric mean of $\mathcal{E}$ is a consequence of the declared summation convention, not a field-moment assumption or a consequence claimed from mean intensity alone.

The summability issue is intrinsic to (3.2): moving one particle so that $\int_I d \neq 0$ makes the averaged energy difference a constant minus a nonzero multiple of the symmetric average in (3.1). Thus finite local-move energy differences for this convention require that limit. Alternative renormalizations must be specified separately.

4. LIFTING FINITE GAUSSIAN PARTICLE KERNELS

The following theorem is independent of Section 3. It is the link between particle-only conditional laws and a stationary field formulation.

Theorem 4.1 (Lifting theorem). *Let P be a stationary law on $\text{Conf}(\mathbb{R})$ of finite positive intensity λ . Suppose that for every bounded deterministic interval $I = (a, b)$ there is a finite real $\mathcal{F}_I$ -measurable variable e_I such that, on $\{N_I = n > 0\}$, the conditional interior law is K_{I,n,e_I} . No condition on e_I on $\{N_I = 0\}$ is needed. Then, conditional on nonemptiness, P admits a measurable deterministic jointly stationary field-DLR lift $E = C(\gamma) + D_\gamma$. The conditional law on nonempty configurations has intensity and almost-sure forward and backward density ρ . Consequently*

$$P(\emptyset) = 1 - \lambda/\rho, \tag{4.1}$$

so necessarily $\lambda \leq \rho$. In particular, if $\lambda = \rho$, the whole law lifts. For every fixed $t \in \mathbb{R}$,

$$C(\theta_t \gamma) = C(\gamma) + D_\gamma(t) \quad P\text{-a.s.} \tag{4.2}$$

on the nonempty component.

Proof. The event $\{\gamma \neq \emptyset\}$ is measurable from $\mathcal{F}_I$ for every I : either the retained exterior is nonempty or $N_I > 0$. Conditioning on this event preserves all the conditional laws and stationarity. Its probability is positive and its conditional intensity is finite. Work on that component until the final paragraph.

Compatibility and identifiability. Let $I = (a, b) \subset J = (c, d)$. Then $\mathcal{F}_J \subset \mathcal{F}_I$. Further conditioning the J -density on the exterior of I fixes the number $k = N_{[c,a]}$ of particles to the left of I . The remaining ordered density in I is Gaussian with coefficient $e_J + k - \rho(a - c)$. This follows by replacing the rank i in the I -block with $k + i$ in the J -block. Disintegration is legitimate because the J -density is strictly positive on its ordered chamber.

Two coefficients e, e' in $K_{I,n,e}$, $n \geq 1$, have likelihood ratio

$$\text{constant} \times \exp \left(2\beta(e - e') \sum_{i=1}^n x_i \right).$$

This ratio can be constant almost everywhere on the chamber only if $e = e'$. Uniqueness of conditional laws therefore gives

$$e_I = e_J + N_{(c,a]} - \rho(a - c) \quad \text{on } \{N_I > 0\}, \quad P\text{-a.s.} \quad (4.3)$$

Measurable patching. Take all rational bounded intervals, together with countably many rational enclosing intervals as needed. Their identities (4.3) hold simultaneously outside one null set. For occupied intervals, (2.2) implies that

$$e_I - D_\gamma(a) \quad (4.4)$$

has the same value for any two choices of I : compare both to a common occupied enclosing interval. Select the first occupied member of the sequence $(-m, m)$, $m \geq 1$, and use (4.4) to define C . This is finite and measurable. Define $E = C + D_\gamma$. For any additional fixed interval, the same identity follows by comparison to the exhausting rational sequence.

The conditioning sigma-fields. Fix $I = (a, b)$, whether or not it is occupied. Take a deterministic increasing sequence of intervals $K_m = (u_m, v_m) \supset I$ exhausting $\mathbb{R}$ and choose its first occupied member K . This choice is $\mathcal{F}_I$ -measurable, since $N_{K_m} = N_I + N_{K_m \setminus I}$. Both e_K and the count $N_{(u_K, a]}$ are $\mathcal{F}_I$ -measurable. Patching gives

$$E(a) = e_K + N_{(u_K, a]} - \rho(a - u_K). \quad (4.5)$$

Every exterior field value follows from this boundary value and the exterior counts, adding the known total N_I when crossing I . Hence the exterior field is $\mathcal{F}_I$ -measurable. Conversely it recovers N_I from $E(b) - E(a) + \rho(b - a)$. It follows that

$$\sigma(\gamma|_{I^c}, E|_{I^c}) = \mathcal{F}_I \quad \text{modulo null sets.} \quad (4.6)$$

Right continuity permits use of countably many field evaluations here; the left boundary value is also recovered by a limit from the exterior. Deterministic endpoints are unoccupied almost surely.

On $N_I > 0$, (4.3) identifies $E(a) = e_I$. On $N_I = 0$ the conditional configuration is empty. Thus (4.6) proves field-DLR. Formula (4.5) also shows that exterior field values are retained under conditional resampling. When $0 \in I$, the new value of C is the value at zero of the bridge (2.4), so it is updated rather than retained.

Stationarity of the lift. For each fixed real t , stationarity of P makes the translated $I+t$ conditional law another version of the I conditional law. Identifiability yields

$$e_I(\theta_t \gamma) = e_{I+t}(\gamma) \quad \text{on } \{N_{I+t} > 0\}, \quad P\text{-a.s.}$$

Compare the countable rational family, its t -translate, and a countable family of enclosing intervals on one full-measure set. Since $D_{\theta_t \gamma}(a) = D_\gamma(a + t) - D_\gamma(t)$, (4.4) gives (4.2). This proves joint stationarity, without asserting an uncountable intersection of null-set complements.

Density and the empty component. Stationarity of the finite-valued field gives

$$\frac{N_{(0,t]}}{t} - \rho = \frac{E(t) - E(0)}{t} \longrightarrow 0 \quad \text{in probability.}$$

Indeed, each of $E(t)$ and $E(0)$ has the same fixed tight distribution. The point-process ergodic theorem at finite intensity gives almost-sure and L^1 convergence of $N_{(0,t]}/t$ to the invariant-component intensity. It must therefore equal ρ . The backward argument is identical. In particular the nonempty component has mean intensity ρ . If $p = P(\emptyset)$ for the original law, its intensity is $\lambda = (1 - p)\rho$, proving (4.1). $\square$

Corollary 4.2 (Application to the particle specification). *Every stationary solution of Definition 3.3 of intensity ρ has the lift $C = -L_\rho(\gamma)$ in Theorem 1.1.*

Proof. The coefficient (3.8) is finite and $\mathcal{F}_I$ -measurable. Apply Theorem 4.1; on any occupied interval, (4.4) equals $-L_\rho(\gamma)$. $\square$

Remark 4.3 (Which other specifications are covered?). The theorem concerns conditional densities, not names of normalization schemes. For $g = -|x|$, an exterior particle potential is affine on I ; the local background has second derivative 2ρ , and the ordered pair energy is linear. Consequently every finite real exterior-potential prescription of this form yields (2.3) and is covered once its measurability and admissibility have been checked. A kernel obtained by mixing distinct unobserved field coefficients need not have this form. Neither such mixtures nor kernels singular with respect to ordered Lebesgue measure are included by assertion.

5. STATIONARY FIELD UNIQUENESS AND QUADRATIC COORDINATES

Theorem 5.1 (Stationary field-DLR uniqueness). *Fix $\beta, \rho > 0$. There is at most one jointly stationary law of (γ, C) of particle intensity ρ with finite-valued $E = C + D_\gamma$ satisfying Definition 2.1. No field-moment assumption is required.*

We give the proof in full in this and the next three sections.

For $a < x_1 < \dots < x_n < b$, write

$$H_{a,b,e}(x) = \int_a^b \left(e + \sum_{i=1}^n \mathbf{1}_{\{x_i \leq t\}} - \rho(t-a) \right)^2 dt.$$

Moving the i th jump gives

$$\begin{aligned} \partial_{x_i} H_{a,b,e} &= (e + i - 1 - \rho(x_i - a))^2 - (e + i - \rho(x_i - a))^2 \\ &= 2\rho(x_i - a) - 2e - 2i + 1. \end{aligned}$$

Consequently, on the ordered chamber,

$$H_{a,b,e}(x) = \rho \sum_{i=1}^n \left(x_i - a - \frac{e + i - \frac{1}{2}}{\rho} \right)^2 + C(a, b, e, n), \quad H''_{a,b,e} = 2\rho I_n. \quad (5.1)$$

No neutrality relation $n = \rho(b-a)$ is required here. The background provides the entire Hessian; the ordered pair interaction is linear.

Set

$$y_i = \rho(x_i - a) - e + \frac{1}{2}, \quad \alpha = \beta/\rho. \quad (5.2)$$

Let $\nu_n = \nu_{n,\alpha}$ be the probability measure on the unbounded ordered chamber $\mathcal{O}_n = \{y_1 < \dots < y_n\}$ with density

$$\nu_n(dy) = Z_n^{-1} \exp \left[-\alpha \sum_{i=1}^n (y_i - i)^2 \right] dy. \quad (5.3)$$

Then the conditional law of y is precisely

$$\mu_{n;A,B} = \nu_n(\cdot \mid W_{A,B}), \quad W_{A,B} = \{A < y_1, y_n < B\}, \quad (5.4)$$

where

$$A = \frac{1}{2} - e, \quad B - n = \frac{1}{2} - f, \quad f = e + n - \rho(b-a). \quad (5.5)$$

For particle endpoints, the same formulas hold with $e = E(a+)$ and $f = E(b-)$, since the endpoints themselves are retained.

There are two useful cancellation identities:

$$y_{i+1} - y_i = \rho(x_{i+1} - x_i), \quad z_i := y_i - i = \frac{1}{2} - E(x_i+). \quad (5.6)$$

Thus the unknown common shift of the centers disappears from physical gaps, while the displacements z_i encode field marks exactly.

6. THE ENTROPY COST OF THE WALLS

Lemma 6.1 (A uniform sublinear wall cost). *For every $\alpha > 0$ and $M < \infty$,*

$$\sup_{|A| \leq M, |B-n| \leq M} \frac{-\log \nu_n(W_{A,B})}{n} \longrightarrow 0. \quad (6.1)$$

More explicitly, for all sufficiently large n depending on α, M ,

$$-\log \nu_n(W_{A,B}) \leq C_\alpha (M + 2 + \sqrt{\log(4n)/\alpha}) n^{2/3}. \quad (6.2)$$

Proof. We first bound the two extreme coordinates. Let σ_c be the law of the ordered values of n independent Gaussians of mean c and variance $(2\alpha)^{-1}$. This law is positively associated: sorting is coordinatewise increasing, and independent variables are associated. On $\mathcal{O}_n$,

$$\frac{d\nu_n}{d\sigma_c}(y) = \text{constant} \times \exp \left(2\alpha \sum_{i=1}^n (i - c)y_i \right).$$

For $c = 1$ the ratio is increasing in every coordinate, so association implies $\sigma_1 \preceq \nu_n$ in stochastic order. For $c = n$ the ratio is decreasing, and $\nu_n \preceq \sigma_n$. The covariance inequalities extend to these integrable exponential ratios by truncation. Gaussian tail bounds and a union bound therefore give, for $R > 0$,

$$\nu_n(y_1 < -R) + \nu_n(y_n > n + R) \leq 2ne^{-\alpha R^2}. \quad (6.3)$$

In particular, with $R = \sqrt{\log(4n)/\alpha}$, the event

$$C_n = \{y_1 \geq -R, y_n \leq n + R\}$$

has probability at least $1/2$.

Put $r = R + M + 2$, $m = \lceil n^{1/3} \rceil$, and $\varepsilon = rn^{-1/3}$. For sufficiently large n , we have $\varepsilon < 1/2$ and $n > 2m + 1$. Define

$$d_i = r \left(1 - \frac{i-1}{m} \right)_+ - r \left(1 - \frac{n-i}{m} \right)_+, \quad T_i(y) = (1 - \varepsilon)y_i + \varepsilon i + d_i.$$

The two supports in d_i are disjoint, and $d_{i+1} - d_i \geq -r/m \geq -\varepsilon$. Hence

$$T_{i+1}(y) - T_i(y) \geq (1 - \varepsilon)(y_{i+1} - y_i) > 0.$$

For $y \in C_n$,

$$T_1(y) \geq M + 2 + \varepsilon(R + 1) > A, \quad T_n(y) \leq n - M - 2 - \varepsilon R < B.$$

Thus $T(C_n) \subset W_{A,B}$. This affine map is injective and has Jacobian $(1 - \varepsilon)^n$.

Writing $q_i = y_i - i$, convexity of the square gives

$$(T_i(y) - i)^2 = ((1 - \varepsilon)q_i + d_i)^2 \leq (1 - \varepsilon)q_i^2 + \frac{d_i^2}{\varepsilon} \leq q_i^2 + \frac{d_i^2}{\varepsilon}.$$

Also $\sum_i d_i^2 \leq 2(m + 1)r^2$. Changing variables on $T(C_n)$ yields

$$\nu_n(W_{A,B}) \geq \frac{1}{2}(1 - \varepsilon)^n \exp \left[-\frac{2\alpha(m + 1)r^2}{\varepsilon} \right].$$

Since $-\log(1 - \varepsilon) \leq 2\varepsilon$,

$$-\log \nu_n(W_{A,B}) \leq \log 2 + 2n\varepsilon + \frac{2\alpha(m + 1)r^2}{\varepsilon} \leq C_\alpha r n^{2/3}.$$

This proves both claims. $\square$

7. GAUSSIAN ENTROPY COMPARISON FOR BLOCK AVERAGES

Write $S(\mu \mid \nu) = \int \log(d\mu/d\nu) d\mu$. The Hessian of the exponent of (5.3) is $2\alpha I_n$. The logarithmic Sobolev inequality on a convex domain and its Herbst consequence imply, for an L -Lipschitz F ,

$$\log \nu_n \left(e^{t(F - \nu_n F)} \right) \leq \frac{t^2 L^2}{4\alpha}, \quad t \in \mathbb{R}.$$

For the concentration and logarithmic Sobolev inequalities, see [3]. They apply here by approximating the indicator of the closed ordered chamber by smooth convex penalties, retaining Hessian at least $2\alpha I_n$, and passing to the limit. The boundary has zero Lebesgue measure; the limiting density need not vanish there. Applying the entropy variational inequality with either sign and optimizing t gives

$$|\mu F - \nu_n F|^2 \leq \frac{L^2}{\alpha} S(\mu \mid \nu_n). \quad (7.1)$$

For a fixed integer $r \geq 1$ and bounded Lipschitz $G : \mathbb{R}^r \rightarrow \mathbb{R}$, define the displacement-mark average

$$F_n(y) = \frac{1}{n-r+1} \sum_{i=1}^{n-r+1} G(y_i - i, \dots, y_{i+r-1} - (i+r-1)). \quad (7.2)$$

Bounded overlap of windows gives the explicit bound $\text{Lip}(F_n)^2 \leq r \text{Lip}(G)^2 / (n-r+1)$, hence $\text{Lip}(F_n)^2 \leq C_{G,r}/n$ for $n \geq 2r$. The same estimate holds for averages of bounded Lipschitz functions of any fixed number of consecutive gaps $y_{i+1} - y_i$.

For the conditioned law in (5.4), exactly

$$S(\mu_{n;A,B} \mid \nu_n) = -\log \nu_n(W_{A,B}). \quad (7.3)$$

Combining (7.1), (7.3), and Lemma 6.1 proves

$$\sup_{|A| \leq M, |B-n| \leq M} |\mu_{n;A,B} F_n - \nu_n F_n| \longrightarrow 0. \quad (7.4)$$

In particular, its square is bounded by $C_{\alpha,G,r}(M+2+\sqrt{\log(4n)/\alpha})n^{-1/3}$ for large n .

Remark 7.1 (Why use entropy directly?). The wall indicator makes a direct relative-Fisher-information comparison with the unbounded reference generally infinite. Equation (7.3) is finite and is the appropriate replacement.

8. STATIONARY RANDOM BLOCKS AND PALM IDENTIFICATION

Let P be a law in Theorem 5.1, and write P^0 for its marked Palm law. First, the density is ρ almost surely in both directions. Indeed, stationarity of the finite-valued field gives

$$\frac{N_{(0,t]}}{t} - \rho = \frac{E(t) - E(0)}{t} \longrightarrow 0 \quad \text{in probability.}$$

The stationary ergodic theorem for the finite-intensity point process gives an almost-sure count-density limit, which must therefore be ρ , even if P is not ergodic. The backward argument is identical. This translation-invariant property passes to Palm measure.

Under P^0 enumerate the points by $\dots < p_{-1} < p_0 = 0 < p_1 < \dots$, put $g_i = p_{i+1} - p_i$, and let T_* be the shift to the successor, translating the field as well. Marked Palm measure is T_* -invariant and $\mathbb{E}_{P^0} g_{-1} = 1/\rho$. For a spatial sample, let a be its first positive particle. The law of the marked environment viewed from a is

$$\tilde{P}^0(d\omega) = \rho g_{-1}(\omega) P^0(d\omega). \quad (8.1)$$

Indeed, Palm inversion over the preceding gap gives $\mathbb{E}_P h(\theta_a \omega) = \rho \mathbb{E}_{P^0}[g_{-1} h]$ for bounded measurable h . This proves the displayed length-bias formula; see [5, Chapter 9] for Palm inversion.

Let b be the $(n+2)$ nd positive particle, so that (a, b) contains exactly n points $x_1 < \dots < x_n$. Retain a, b , every other exterior particle, and the field outside (a, b) .

Lemma 8.1 (Random-block disintegration and tight walls). *Conditionally on this retained information, the transformed interior coordinates (5.2), with $e = E(a+)$, have law $\mu_{n;A,B}$ in (5.4). The family $(A, B - n)$ is tight as $n \rightarrow \infty$.*

Proof. For disintegration, first restrict to $b < R$ for a deterministic integer R . Apply field-DLR in $(0, R)$ and condition further on every ordered coordinate except the n middle particles, retaining the resulting exterior field. Moving these middle particles within (a, b) preserves the selected ranks. The part of $\int_0^R E^2$ outside (a, b) is fixed. Thus the remaining density is (2.4) on (a, b) . The events $\{b < R\}$ are retained-information measurable and exhaust probability one. Equations (5.1)–(5.5) identify the law.

To prove tightness, let g_* be the entire gap containing the spatial origin. On $\{g_* \leq L\}$, the density in (8.1) is bounded by ρL . Invariance under T_* gives, uniformly in n ,

$$\begin{aligned} P(|E(a+)| > R \text{ or } |E(b-)| > R) \\ \leq P(g_* > L) + \rho L \left\{ P^0(|E(0+)| > R) + P^0(|E(0-)| > R) \right\}. \end{aligned}$$

The Palm field values are finite almost surely. First choose L large, then R large. Formula (5.5) proves the claim. No Palm moment of the field is being assumed. $\square$

Proof of Theorem 5.1. Fix r and a bounded Lipschitz G as in (7.2). By (5.6), the physical block statistic is

$$\mathfrak{F}_n(\gamma, E) = \frac{1}{n - r + 1} \sum_{i=1}^{n-r+1} G\left(\frac{1}{2} - E(x_i+), \dots, \frac{1}{2} - E(x_{i+r-1}+)\right).$$

Its conditional expectation is $\mu_{n;A,B} F_n$. Lemma 8.1, (7.4), and boundedness give

$$\mathbb{E}_P \mathfrak{F}_n - \nu_n F_n \longrightarrow 0. \quad (8.2)$$

Explicitly, restrict first to $|A|, |B - n| \leq M$, take $n \rightarrow \infty$, and then let $M \rightarrow \infty$ using uniform tightness.

We identify the limit on the left via marked Palm measure. Let $\mathcal{I}$ be the T_* -invariant sigma-algebra. Birkhoff's theorem for the bounded mark observable G yields the conditional limit $\mathbb{E}_{P^0}[G \mid \mathcal{I}]$ for the successive block averages. The derived spatial density implies $p_m/m \rightarrow 1/\rho$ under Palm measure. Birkhoff applied to the integrable gap therefore gives

$$\mathbb{E}_{P^0}[g_{-1} \mid \mathcal{I}] = 1/\rho.$$

Use (8.1), dominated convergence with weight ρg_{-1} , and conditioning on $\mathcal{I}$. It follows that

$$\mathbb{E}_P \mathfrak{F}_n \longrightarrow \mathbb{E}_{P^0} G\left(\frac{1}{2} - E(p_0+), \dots, \frac{1}{2} - E(p_{r-1}+)\right). \quad (8.3)$$

The finitely shifted starting index does not change this limit.

For any second candidate Q , the same deterministic $\nu_n F_n$ occurs in (8.2). Equations (8.2)–(8.3) therefore identify all finite consecutive distributions of $z_i = \frac{1}{2} - E(p_i+)$ under P^0 and Q^0 . Successor invariance identifies every finite window with arbitrary integer indices.

The entire marked Palm configuration is determined by this sequence:

$$\rho g_i = 1 + z_{i+1} - z_i, \quad p_0 = 0, \quad E(p_i+) = \frac{1}{2} - z_i, \quad (8.4)$$

and the field has slope $-\rho$ between particles. Thus the marked Palm laws coincide. Marked Palm inversion, for bounded measurable h ,

$$\mathbb{E}_P h = \rho \mathbb{E}_{P^0} \int_0^{g_0} h(\theta_t(\gamma, E)) dt,$$

now proves $P = Q$. $\square$

9. EXISTENCE FOR THE SPECIFIED NORMALIZATION

Only existence uses estimates from the classical finite-volume construction. The preceding lifting and uniqueness arguments apply to every candidate in their stated classes.

For $N \in \mathbb{N}$, let P_N be the law of $2N$ ordered particles in $B_N = (-N/\rho, N/\rho)$, with density proportional to

$$\exp \left[-\beta \int_{B_N} E_N(x)^2 dx \right], \quad E_N(x) = N_{(-N/\rho, x]} - \rho(x + N/\rho), \quad x \in B_N.$$

The field has value zero at both endpoints. Extend it to the whole line with derivative $\gamma_N - \rho dx$, where γ_N has no particles outside B_N . This extension is used only to put all laws on one state space; finite-volume energies are integrated over B_N .

Lemma 9.1 (Imported finite-volume estimates). *For fixed $\beta, \rho > 0$, there are positive constants A, B such that, uniformly in N and $x \in B_N$,*

$$P_N(|E_N(x)| > u) \leq Ae^{-Bu^3}, \quad u \geq 0. \quad (9.1)$$

For every $\delta > 0$, there are $A_\delta, B_\delta > 0$ such that, uniformly over intervals $J \subset B_N$,

$$P_N \left(\left| \frac{1}{|J|} \int_J E_N(x) dx \right| > \delta \right) \leq A_\delta e^{-B_\delta |J|}. \quad (9.2)$$

These are the strictly one-dimensional estimates of [2, Theorem 6.2 and Proposition 8.1]; the latter is the one-dimensional proof of their volume-average bound, Theorem 3.4. Their unit-density convention is $U = \frac{1}{2} \int K^2$. The conversion is $y = \rho x$, $K(y) = -E_N(y/\rho)$, and $\beta_{\text{AJJ}} = 2\beta/\rho$. No numerical value of an exponential constant is needed here. The displayed bounds at arbitrary spatial endpoints follow from the bounds on the density grid: on each grid cell, the field is bounded in absolute value by one plus the larger absolute endpoint value. Rounding the endpoints of J changes its integral only by two such cell integrals; (9.1) controls those errors. Thus the constants are uniform in the box and the interval.

Proposition 9.2. *There exists a jointly stationary field-DLR law of intensity ρ . It has $\mathbb{E}|E(0)| < \infty$ and almost surely*

$$\lim_{R \rightarrow \infty} \frac{1}{2R} \int_{-R}^R E(x) dx = 0. \quad (9.3)$$

Its particle marginal satisfies Definition 3.3. The field has finite moments of every finite order.

Proof. Independently choose U_N uniformly in $[-N/(2\rho), N/(2\rho)]$ and let $\bar{Q}_N$ be the law of $(\theta_{U_N} \gamma_N, E_N(\cdot + U_N))$. Use vague convergence for locally finite integer-valued measures (temporarily allowing multiple points), and L^1_{loc} convergence for fields. Translations are continuous in this topology. Changing U_N to $U_N + t$ changes its law in total variation by $O(\rho|t|/N)$. Consequently every subsequential limit is stationary.

Here is the required compactness argument. For every fixed compact interval K , all its shifted points lie inside B_N for large N . Equation (9.1) bounds the expectation of the field's $L^1(K)$ norm uniformly. For fixed $a < b$ in that region,

$$N_{(a,b]} = E_N(b) - E_N(a) + \rho(b - a).$$

The same estimate therefore bounds the expected number of particles on K . Since the distributional derivative is $\gamma_N - \rho dx$, this also bounds the expected total variation on K . Compactness of bounded sets of $BV(K)$ in $L^1(K)$, followed by a diagonal argument and the corresponding vague compactness for counting measures, proves tightness. Choose a limit Q . The distributional identity passes to the limit, so $E' = \gamma - \rho dx$. The field admits a unique right-continuous representative and is finite-valued. The limit counting measure remains integer-valued, with possible multiplicities at this stage.

Stationarity and the count bounds imply that its mean counting measure is a finite constant times Lebesgue measure. In particular, no deterministic point is an atom almost surely. The local L^1 bound passes to the limit; stationarity then gives $\mathbb{E}_Q|E(0)| < \infty$. Taking expectations in the charge identity proves $\mathbb{E}_Q N_{(a,b]} = \rho(b - a)$.

We next pass the conditional equations, spelling out the continuity point that prevents a hidden simplicity assumption. For a fixed deterministic $I = (a, b)$ let Γ_I replace its particles by the Gaussian density in Definition 2.1 and retain its exterior field. Define this kernel also for integer-valued incoming measures with multiple points. If the incoming measure has no atoms at a, b , its interior count is locally constant under vague convergence, and its boundary field value converges under the joint topology above. The latter assertion follows from $E' = \gamma - \rho dx$: convergence of monotone primitives holds at their continuity points. For each fixed count, the ordered Gaussian density depends continuously on the finite boundary value, by dominated convergence on the bounded chamber. Hence $\Gamma_I f$ is continuous there for every bounded continuous f on the joint state space. Resampled fields also converge in L^1_{loc} .

For large N , every shifted box contains I , and the finite-volume conditional equation holds for each origin in the averaging range. Thus $\overline{Q}_N \Gamma_I = \overline{Q}_N$. The almost-sure continuity just established gives $Q \Gamma_I = Q$. As Γ_I retains the exterior information, this identity is equivalent to the conditional field-DLR equation. Moreover, Γ_I produces distinct interior points with probability one. Applying this to a countable covering of the line proves that γ is simple almost surely. The boundary-evaluation continuity also transfers (9.1) to the limiting field at each deterministic point (with an arbitrarily smaller threshold before passage to the limit). Thus the limiting field has finite moments of every finite order.

For each fixed R , (9.2) passes to the averaged laws and their limit, because integration on $[-R, R]$ is continuous in L^1_{loc} . It bounds the probability that the absolute average exceeds δ by $A_\delta e^{-2B_\delta R}$. Borel–Cantelli gives zero limiting average along integer R . The stationary ergodic theorem, applicable because $\mathbb{E}_Q|E(0)| < \infty$, gives an almost-sure limit for the full real parameter; that limit must be zero. This proves (9.3) without asserting ergodicity.

Finally, writing $C = E(0)$ gives $D_\gamma(x) = E(x) - C$. Equation (9.3) therefore gives the Cesàro limit in (3.1), with $L_\rho(\gamma) = -C$. The particle law is supported on the declared admissibility domain. Canonical replacement leaves the reconstructed exterior field unchanged, so that exterior particles and their retained interior count determine the boundary value. Its field-DLR kernel is thus exactly the particle kernel of Definition 3.3. $\square$

Combining Proposition 9.2 with Theorem 5.1 and the lifting theorem completes existence and uniqueness for the particle specification. The moments and zero-mean property used in this section were obtained for the constructed law; they were not hypotheses on the competing laws in the uniqueness theorem.

10. CONSEQUENCES AND RELATION TO THE LITERATURE

Proof of Theorem 1.1. Corollary 4.2 maps every candidate to a jointly stationary field-DLR law. Theorem 5.1 identifies all such lifts and hence their particle marginals. Proposition 9.2 supplies a candidate for the declared summation convention. The almost-sure density statement is part of Theorem 4.1. $\square$

Corollary 10.1 (All finite-intensity stationary Gaussian-kernel laws). *Let P be stationary and satisfy the conditional-density hypothesis of Theorem 4.1 with finite intensity $\lambda \geq 0$. Then $0 \leq \lambda \leq \rho$ and*

$$P = \frac{\lambda}{\rho} P_{\beta, \rho} + \left(1 - \frac{\lambda}{\rho}\right) \delta_\emptyset.$$

Proof. For $\lambda > 0$, Theorem 4.1 determines the empty mass, and Theorem 5.1 identifies the lifted nonempty component with the constructed law. A stationary zero-intensity counting measure is empty almost surely. $\square$

This corollary underscores why mean intensity cannot simply be replaced by an unproved almost-sure density assertion. The extra canonical vacuum component is a real possibility when intensity is not fixed to the background. Conversely, no random conditional distribution of the additive field constant is needed for the particle laws treated here: their finite Gaussian conditional coefficients already determine it.

For the unique law, the moment estimates imported only for existence now hold automatically. In particular, the construction and uniqueness give $\mathbb{E}E(0)^2 < \infty$ and

$$\text{Var } N_{(0,t]} = \mathbb{E}(E(t) - E(0))^2 \leq 4\mathbb{E}E(0)^2.$$

These are conclusions, not requirements placed on competitors.

Kunz’s transfer-operator analysis [4] and the field viewpoint of Aizenman–Martin [1] are classical foundations of this model. The published overview [6, Section VI.A.2.e] describes the periodic crystal states, their uniform phase average, and the renormalized canonical equations. We do not use that overview’s classification statement as a theorem covering the present candidate class: the summation convention and the conditional lifting argument are supplied explicitly above. Nor do we assign an unverified exact admissibility class to either of the original classical papers.

The reconstruction discussion in [2, Section 5, equations (30)–(32) and the ensuing remarks] uses estimates for finite-volume limits. It does not by itself establish those estimates for every arbitrary particle-DLR solution. Theorem 4.1 supplies that missing logical implication here through nested conditional densities.

Crystalline phase multiplicity is compatible with the theorem. A fixed-phase law need not be invariant under all real translations; a uniform average over a periodic phase family is stationary. No classification of all nonstationary phases was used at any step of the uniqueness proof. The scope is precisely the finite-coefficient conditional-density class of Theorem 4.1, and, for the existence-and-uniqueness assertion, the particle summation convention of Definition 3.3.

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